BanasthaliVidyapith
Department of Computer Science
Course Handout: B.Tech.(IT) VII
July – December 2026

Date: 24 June 2026

Course Code: IT 401 **Course Name: Data Mining and Warehousing**

Credit Points:4 **Max. Marks: 100(CA: 40 + ESA: 60)**

Course Instructor: Dr. Bharti Nathani, Assistant Professor, Computer Science, B.Tech. (IT) VII Semester

Course Outcomes:

On completion of the course, the student will be able to:

CO 1 Describe the basic concepts of data mining and data warehousing.

CO 2 Apply skills for different data mining techniques in the analysis of data.

CO 3 Describe the schema and organization of data in data warehouse.

CO 4 Apply data mining techniques in wide variety of data like spatial, time series, text, multimedia and World Wide Web.

CO 5 Get experience and encouragement of exploring data mining techniques in real world and doing research.

Syllabus

Section A

Introduction to data mining - Definition, why data mining, on what kind of data, knowledge discovery in databases (KDD), appropriateness of data mining, expert systems vs. data mining systems, data mining functionalities, classification of data mining systems, data mining techniques, major issues. Data Preprocessing - data cleaning, integration and transformation, data reduction, data discretization and concept hierarchy generation.

Section B

Association rules - Market Basket Analysis, Apriori algorithm for generating association rules, FP-growth algorithm. Classification and prediction - decision tree induction, other methods for classification (introduction only), Prediction - linear and nonlinear regression, Evaluating the accuracy of a classifier or predictor. Clustering - Categorization of Major cluster methods, the K-means algorithm, hierarchical clustering (agglomerative and division method).

Section C

Data Warehouse - difference between operational database and warehouse, data warehouse design, three tier architecture, online analytical processing (OLAP), data warehouse implementation. Trends in data mining – system products and research prototypes. Social impacts of data mining. Applications of data mining. Brief overview of mining time series databases, spatial databases and World Wide Web mining.

Suggested Books:

R1. Han Jiawei, Kamber Micheline. *Data Mining: Concepts and Techniques*, 2nd ed. New Delhi: Morgan Kaufmann Publishers, 2006.

R2. Roiger Richard J., Geatz Michael W. *Data Mining: A tutorial-based Primer*, Delhi: Pearson Education, Inc., 2003

R3. Mattison Rob. *Web Warehousing and Knowledge Management*, New Delhi: Tata McGraw Hill Education Private Limited, 1999

R4. Berson Alex, Smith Stephen J. *Data Warehousing, Data Mining & OLAP*, New Delhi: Tata McGraw Hill Education Private Limited, 1997

R5. Inmon William H. *Building the Data Warehouse*, 4th ed. New Delhi: Wiley Publishing Inc., 2005

Suggested E-Learning Material:

E1. Data Mining: https://nptel.ac.in/courses/106105174/

E2. Data Mining: Concepts and Techniques: https://hanj.cs.illinois.edu/bk3/bk3_slidesindex.htm

Assessment Schedule:

Component	Mark	Submission/ Examination date	Allotment
Home assignment I**	10	27 August, 2026	Topics shall be allotted in the class by 5 August 2026
Periodical test I	10	9-12 September, 2026*	Lecture No. 01 to 33
Home assignment II**	10	7 October, 2026	Topics shall be allotted in the class by 14 September, 2026
Periodical test II	10	30 October, 2026 to 2 November, 2026*	Lecture No. 34 to 47
Semester Examination	60	2-19 December, 2026*	Lecture No. 01 to 50 (Entire Syllabus)

* Subject to change.

** Assignment marks will be based on written document(s)/ any other component(s) as decided by the instructor(s).

Lecture-Wise Schedule:

Lec. No.	Topics to be covered	Text/ Ref. Books
1-4	**Introduction to Data Mining, Need of Data Mining, Kind of Data that can be Mined, Knowledge Discovery in Databases (KDD), Appropriateness of Data Mining, Expert Systems v/s Data Mining Systems, Data Mining Functionalities, Classification of Data Mining Systems, Data Mining Techniques, Major Issues in Data Mining**	**R1/ R3/ R5**
5-12	**Data Cleaning, Data Integration and Transformation, Data Reduction, Data Discretization and Concept Hierarchy Generation**	**R1/ R5**
13-15	**Data Generalization and Summarization, Analytical Characterization**	**R1/ R3**
16-20	**Association Rules, Market Basket Analysis, Apriori Algorithm for Generating Association Rules, FP-Growth Algorithm**	**R1/ R2/ R3**
21-26	**Introduction to Classification, Classification by Decision Tree Induction, Introduction to Other Classification Methods**	**R1/ R2/ R3**
37-29	**Prediction through Linear and Nonlinear Regression, Evaluating the Accuracy of a Classifier or Predictor**	**R1/ R2/ R3/**
30-33	**Introduction to Clustering, K-Means Algorithm, Agglomerative and Divisive Hierarchical Clustering**	**R1/ R2/ R4**
34-37	**Introduction to Data Warehouse, Differences between Operational Database Systems and Data Warehouses, Trends and Planning of Data Warehousing, Defining Business Requirements, Three-Tier Data Warehouse Architecture**	**R1/ R5**
38-43	**Data Modeling Strategy, Fact Tables, Dimension Tables, Schema: Star, Snowflake, Constellation, Multidimensional Data Models, Data Cube, On-Line Analytical Processing (OLAP) Models, OLAP Operations, Data Warehouse Implementation**	**R1/ R5**
44-47	**Data Mining System Products and Research Prototypes, Social Impacts of Data Mining, Data Mining Applications**	**R1/R2/ R4**
48-50	**Brief Overview of Mining Time-Series Data, Spatial Data Mining and Mining the World Wide Web**	**R1/R2/ R3**

Dr. **Bharti Nathani**